OxyProof

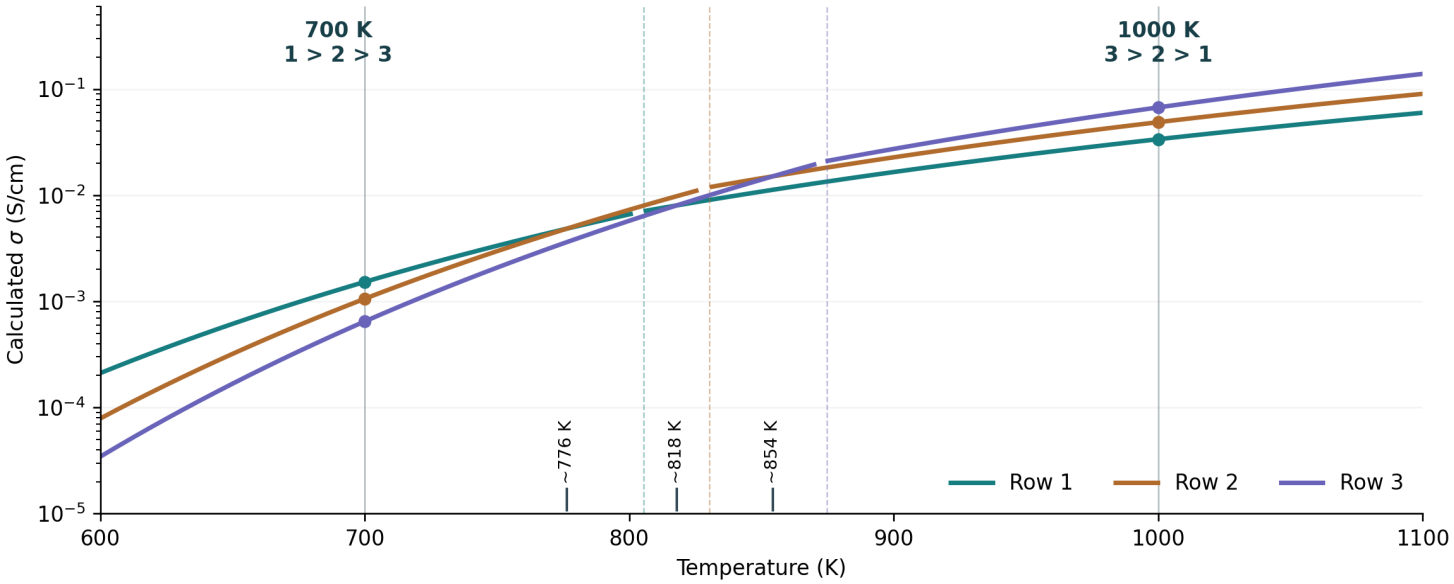
Temperature-Aware, Evidence-Grounded Materials Decision Agent

When the best material changes, can your AI explain why?
Published evidence + bounded AI tools + deterministic physics + independent auditing.

THE RESEARCH DECISION

A shortlist at one operating temperature can hide a different choice across an operating range. OxyProof makes the ranking, its fitted model and the missing evidence inspectable.

FLAGSHIP RESULT | CONDITIONAL MODEL RANKING REVERSAL



Same curated source group: bulk, two-probe AC, 600-1100 K; Huang et al. (1998), Table 3 as curated by Jang et al. Solid branches are fitted-model calculations; dashed lines are curated T* boundaries. Exact T* is withheld.

Rows 1 / 2 / 3: La0.95Sr0.05Ga0.95Mg0.05O2.95 / La0.95Sr0.05Ga0.9Mg0.1O2.925 / La0.95Sr0.05Ga0.85Mg0.15O2.9. Analytical intersections: approximately **776, 818, 854 K**.

Model-derived intersections, not measured crossovers. Uncertainty bounds unavailable.

FOUR LAYERS THAT KEEP AI CLAIMS BOUNDED

1. Plan: DeepSeek selects an allowlisted scientific tool. **2. Validate:** independent guards retain constraints and reject unsupported requirements. **3. Calculate:** sigma = A/T exp[-Ea/(kB T)]; Ea meV to eV, A K S/cm, T K, output S/cm. **4. Audit:** conditional ordering, source rows, regimes, warnings and code fingerprints survive JSON export.

EVIDENCE | RECORDED RUNS, BOUNDED CLAIMS

3/3 genuine DeepSeek scenarios (21 Oct 2026): screening, interval analysis and constraint abstention. Video shows labelled replay; no new inference. **49/49 targeted integrity cases** and **21 Chromium E2E tests** reported passed in retained artifacts. These selected checks are not general scientific accuracy. Full pinned dataset: **483 records**; offline analysis needs no model service.

WHERE THE EVIDENCE STOPS

Original row-specific atmosphere, preparation, density and full methods/Table 3 remain unverified. No validated fit uncertainties, humidity tolerance or 10,000-hour lifetime. Unsupported conditions cause abstention. No new material, experiment or industrial performance is claimed.

Code: github.com/miku-wwl/oxyproof-osi-2026
Data: Jang et al. (2026), Zenodo v1, CC BY 4.0: 10.5281/zenodo.18947543
Method: Scientific Data: s41597-026-07100-x | **Original source:** 10.1111/j.1151-2916.1998.tb02662.x
Video: [PUBLIC VIDEO URL - owner upload pending]. Sources verified 20 Oct; review 21 Oct 2026.
Public repository available. New submission assets are local. Competition entry not submitted.